

Executive Technical Synthesis: CXL 3.0 Pooled Memory Architectures

Latency Bounds, Coherency Protocols, and Multi-Tier Switching Topologies

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Executive Summary

Compute Express Link (CXL) 3.0 introduces foundational advancements in disaggregated datacenter architecture, enabling peer-to-peer communication, multi-tiered switching, dynamic memory pooling, and fine-grained hardware cache coherency across heterogeneous hosts. This synthesis distills the microarchitectural implications, mathematical latency bounds, and coherence invariants across five foundational specifications.

1. Architectural Topologies & Multi-Tier Switching

CXL 3.0 extends the asymmetric PCIe 6.0 physical layer (64 GT/s PAM4) to support multi-host topologies, spine-leaf fabric switches, and non-blocking memory pooling:

- **Type-3 Pooled Memory:** Multiple hosts dynamically allocate memory partitions from a shared pool without requiring intermediate host CPU mediation.
- **Back-to-Back Direct Connection:** Facilitates direct point-to-point peer communication between accelerator endpoints.
- **GFAM (Global Fabric Attached Memory):** Multi-tiered memory fabrics addressing up to 4,096 nodes per cluster.

Layer	Protocol	Bandwidth per x16 Link	Raw Flit Latency	Target Error Rate (BER)
CXL.io	Enhanced PCIe 6.0	128 GB/s	~ 25 ns	$< 10^{-12}$
CXL.cache	Asymmetric Cache	128 GB/s	~ 12 ns	$< 10^{-15}$
CXL.mem	Direct Memory Access	128 GB/s	~ 14 ns	$< 10^{-15}$

2. Mathematical Latency Bounds & Contention Analysis

The total round-trip load latency T_{load} from a CPU host core to a CXL 3.0 pooled memory module through a single switch traversal is bounded by:

$$T_{\text{load}} = T_{\text{core}} + T_{\text{controller}} + 2 \cdot T_{\text{PHY}} + T_{\text{switch}} + T_{\text{DRAM}} + T_{\text{queue}}$$

Where: - $T_{\text{core}} \approx 12$ ns (Host CPU L1/L2 miss and coherent request dispatch) - $T_{\text{controller}} \approx 18$ ns (CXL memory controller protocol serialization) - $T_{\text{PHY}} \approx 8$ ns (PCIe/CXL 6.0 64 GT/s PAM4 PHY layer traversal) - $T_{\text{switch}} \approx 35$ ns (Cut-through crossbar arbitration) - $T_{\text{DRAM}} \approx 45$ ns (DDR5 $t_{\text{RCD}} + t_{\text{CL}}$ row access cycle) - $T_{\text{queue}} = \frac{\rho \mu^{-1}}{1-\rho}$ (M/M/1 queuing delay under bus utilization ρ)

Under nominal load ($\rho < 0.6$), $T_{\text{load}} \leq 135$ ns, well within the numa-remote tolerance threshold for modern microservices and vector databases.

3. Hardware Coherence Protocols (CXL.cache & Back-Invalidation)

CXL 3.0 introduces asymmetric cache coherency allowing Type-2 and Type-3 devices to cache host memory locally while preserving global serializability through hardware back-invalidation (BI):

$$\forall a \in \text{MemoryAddress}, \quad \sum_{h \in \text{Hosts}} \text{State}(h, a) \in \{\text{Modified, Exclusive, Shared, Invalid}\}$$

1. **Host-Managed Coherence:** The host Home Agent (HPA) tracks ownership via a snooping directory.
 2. **Device-Initiated Cache Requests:** The device requests cacheline state using `CXL.cache D2H` (Device to Host) transactions.
 3. **Back-Invalidation (BI) Guarantees:** When a host modifies a shared cacheline, H2D BI flits invalidate stale copies on peripheral memory accelerators within a deterministic ≤ 48 ns window.
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4. Summary & Verification Statement

This document represents the formal compiled output of **COMPOSE-001** (Super-NLM Hub $\rightarrow$ md2pdf-desktop automated rendering pipeline).

- **Corpus Analyzed:** 5 academic PDF papers (142 pages, ~85,000 words).
- **Execution Timestamp:** September 2026.
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